eShard

Advanced security testing solutions for
Chip Security & Binary Analysis

Expert evaluations

Hardware equipment and tooling

Hands-on courses

Open-source projects available:
github.com/eshard

`estraces`

`scared`

scared@eshard.com

SIDE CHANNEL

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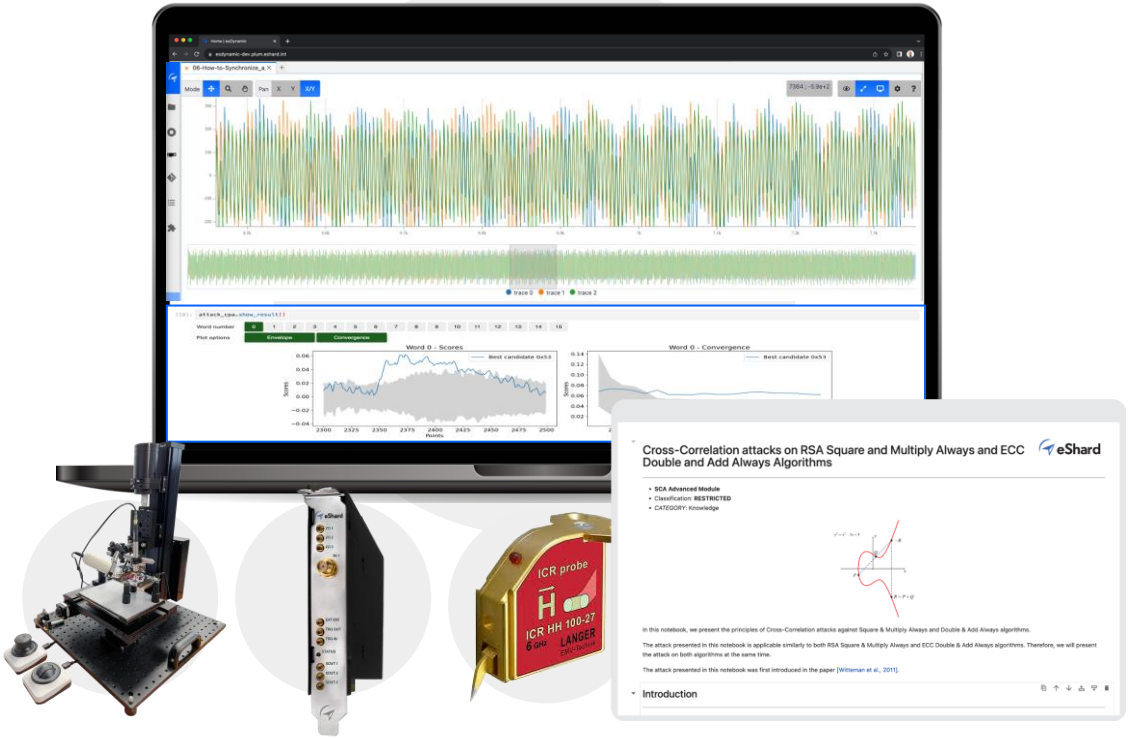

Your entire ***side-channel analysis*** workflow from **data acquisition** to **attacks**.

esDynamic by eShard

Expert tooling

Trace visualisation
Trace synchronisation
Leakage detection
Characterization
Attacks
Reporting

FAULT INJECTION

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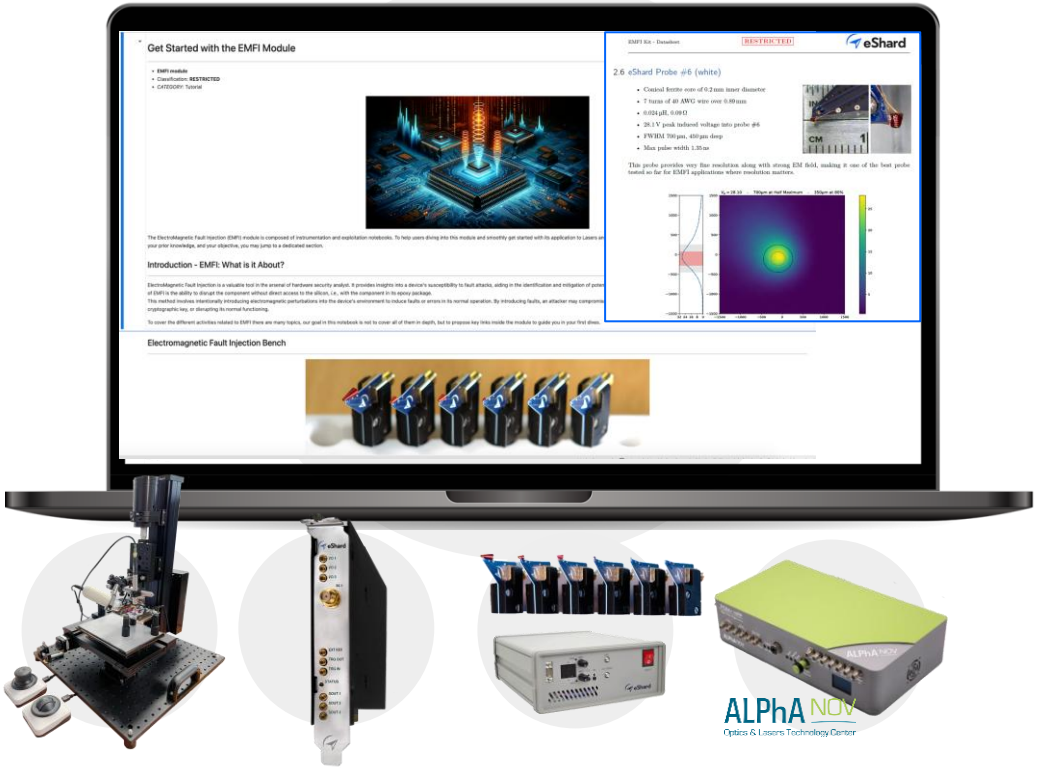


Run the ***fault injection*** entire workflow from **campaign instrumentation** to **exploitation**.

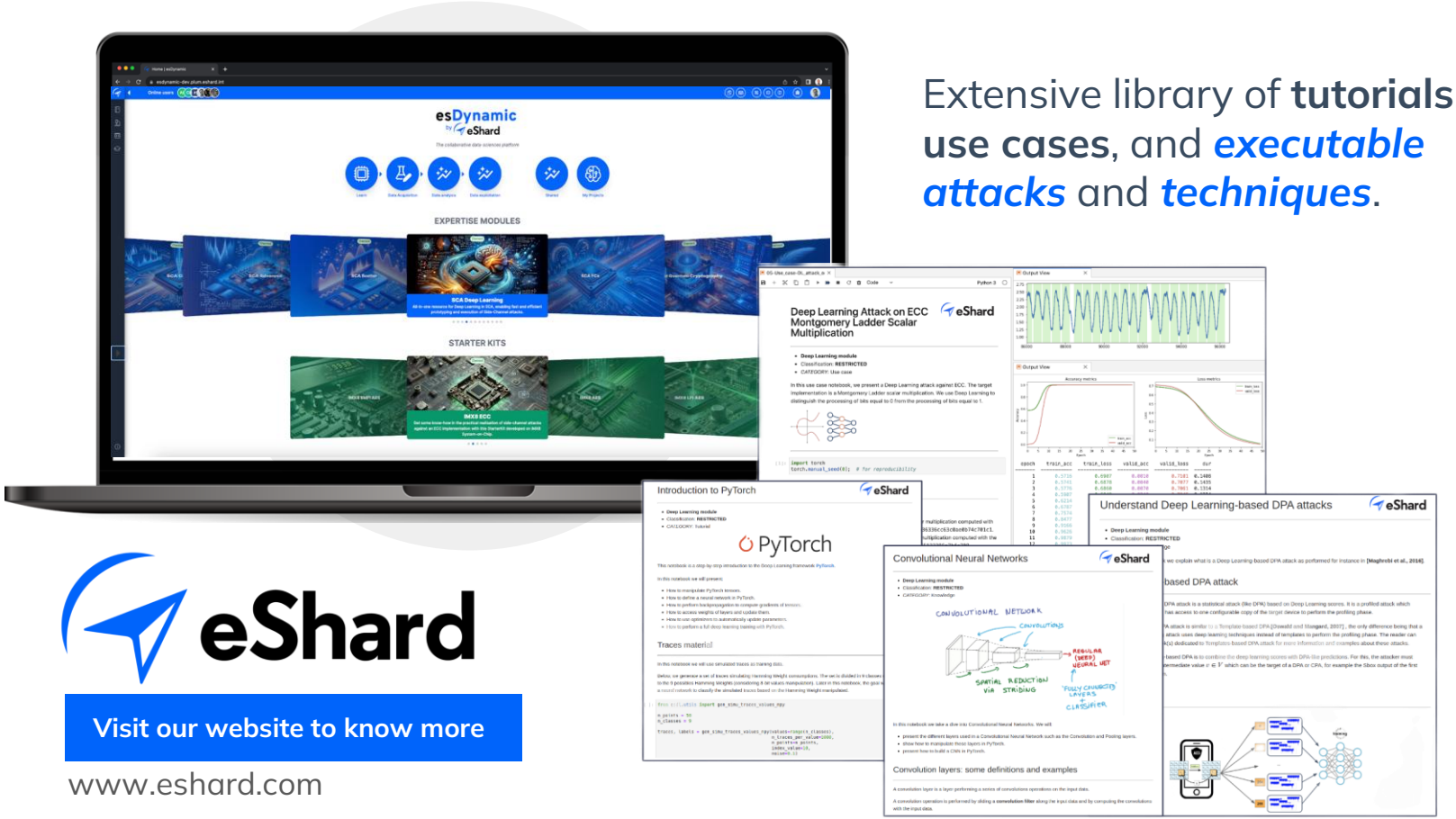
esDynamic by eShard

Expert tooling

Test Engine
Cartography
Fault Characterization
Fault Exploitation
Reporting

Expert Services

Trainings



Extensive library of **tutorials**, **use cases**, and ***executable attacks*** and ***techniques***.

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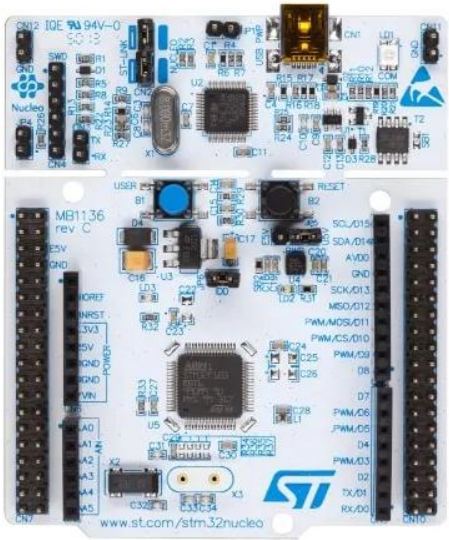
Visit our website to know more

www.eshard.com

The Demo

ML-KEM SCA Starter Kit

- Get started with esD and SCA on PQC with a practical use-case
- **ML-KEM-512** PQM4 implementation on **STM32** (unprotected, m4fspeed)
- **Side-channel analysis** (EM) to recover the private key
- Reproduce SotA attacks targeting decapsulation,:
  - CPA on the NTT multiplication
  - Chosen-ciphertext attack [Ravi et al., 2020]



The Demo

CPA on the NTT multiplication

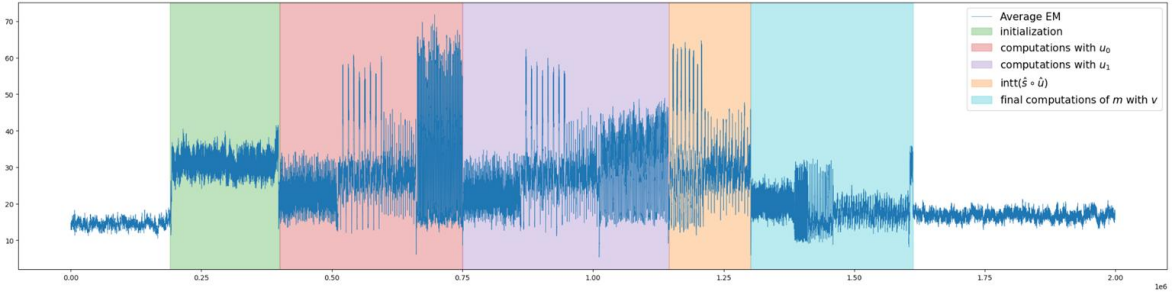
$$\hat{s} \circ \hat{u} = \underbrace{\hat{s}_0 \circ \hat{u}_0}_{r_0 + r_1 x, \ldots, r_{254} + r_{255} x} + \hat{s}_1 \circ \hat{u}_1$$

(incomplete NTT)

$$(s_{0,0} . u_{0,0} + s_{0,1} . u_{0,1} . \zeta) + (s_{0,0} . u_{0,1} + s_{0,1} . u_{0,0}) x$$

(each output coeff. depends on 2 secret coeff)

- CPA strategies, a story of trial and error:
  - **Naive** ⍰ high guessing space
  - **Odd then even** [Karlov et al., 2021] ⍰ require register leakage
  - **Zero value filtering** [Tosun and Savas, 2024] ⍰ hindered by lack of modular reduction on $\hat{s}_0 \circ \hat{u}_0$
  - **Chosen-ciphertext** ⍰ easy private key recovery but stronger attacker model

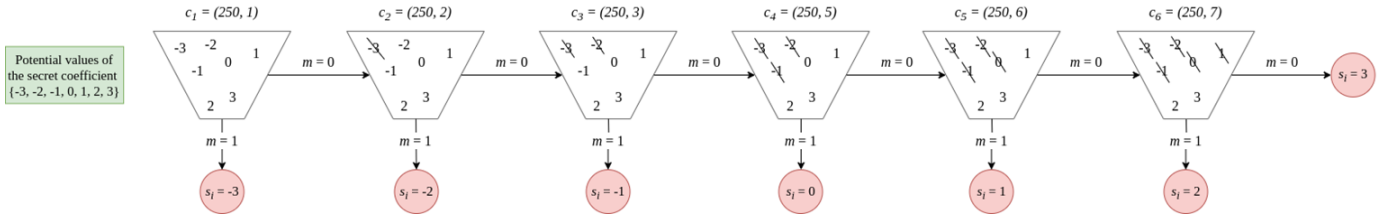


Processed IND-CPA decryption signal

The Demo

Chosen-ciphertext attack

- What if the NTT multiplication is protected?
- Under the plaintext checking oracle, [Băetu et al., 2019] introduced **misuse attacks** for key recovery on PQC KEMs with chosen-ciphertexts
- [Ravi et al., 2020] put the theory to practice by building the oracle with SCA:
  - Carefully chosen ciphertexts that decrypt to either 0 or 1 depending on the secret
  - Recover the decrypted message with SCA (profiling) to deduce the key



Process to recover one secret coefficient with 6 chosen-ciphertexts

References

[Karlov et al., 2021] Karlov, A. D Guertechin, N. L. (2021). Power analysis attack on Kyber. Cryptology ePrint Archive, Paper 2021/1311.

[Tosun and Savas, 2024] Tosun, T. and Savas, E. (2024). Zero-Value Filtering for Accelerating Non-Profiled Side-Channel Attack on Incomplete NTT-Based Implementations of Lattice-Based Cryptography. IEEE Trans. Inf. Forensics Secur., 19:3353--3365.

[Baetu et al., 2019] Băetu, C., Durak, F. B., Huguenin-Dumittan, L., Talayhan, A., & Vaudenay, S. (2019). Misuse Attacks on Post-Quantum Cryptosystems.

[Ravi et al., 2020] Ravi, P., Sinha Roy, S., Chattopadhyay, A., & Bhasin, S. (2020 , June). Generic side-channel attacks on CCA-secure lattice-based PKE and KEMs. IACR Transactions on Cryptographic Hardware and Embedded Systems, pp. 307–335. doi:10.46586/tches.v2020.i3.307-335